Parameter Binding in Web API

Overview



Parameter in Web API

Default Parameter

Primitive Type Parameter

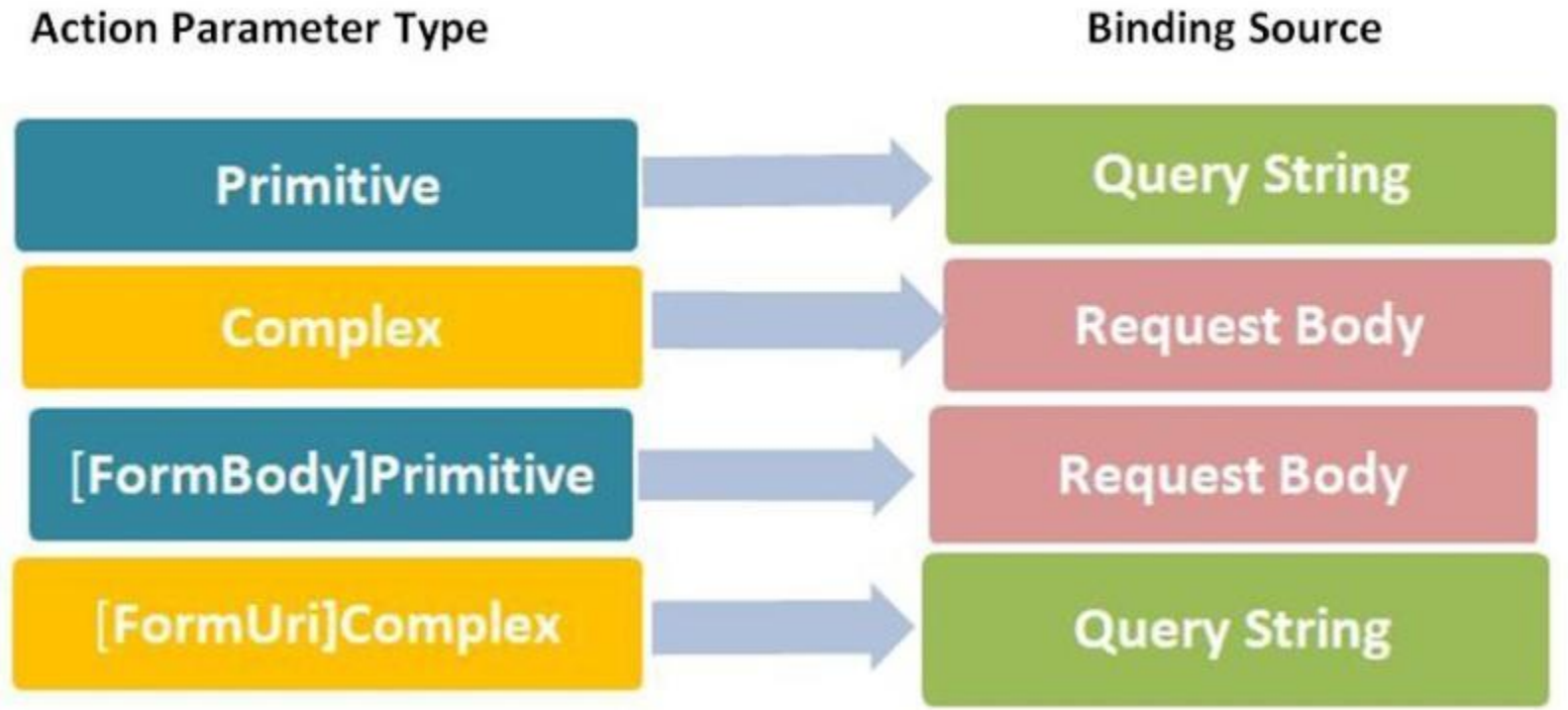
Complex Type Parameter

[FromUri] and [FromBody]

Parameter in Web API

- **Parameter Binding:**

By default, if parameter type is of .NET primitive type such as int, bool, double, string, GUID, DateTime, decimal or any other type that can be converted from string type then it sets the value of a parameter from the query string



Default Parameter

HTTP Method	Query String	Request Body
GET	Primitive Type, Complex Type	NA
POST	Primitive Type	Complex Type
PUT	Primitive Type	Complex Type
PATCH	Primitive Type	Complex Type
DELETE	Primitive Type, Complex Type	NA

Primitive Type Parameter

Example: Primitive Parameter Binding

```
[ApiController]
public class StudentController : ControllerBase {
 public Student Get(int id) { }
 }
```

```
http://localhost/api/student?id=1
```

```
[ApiController]

public class StudentController : ControllerBase
{
public Student Get(int id, string name)
{
}
}
```

```
http://localhost/api/student?id=1&name=steve
```

Example: Primitive Parameter Binding through Request Body

```
[ApiController]
public class StudentController : ControllerBase {
 public Student Get([FromBody] int id) { }
 }
```

http://localhost/api/student

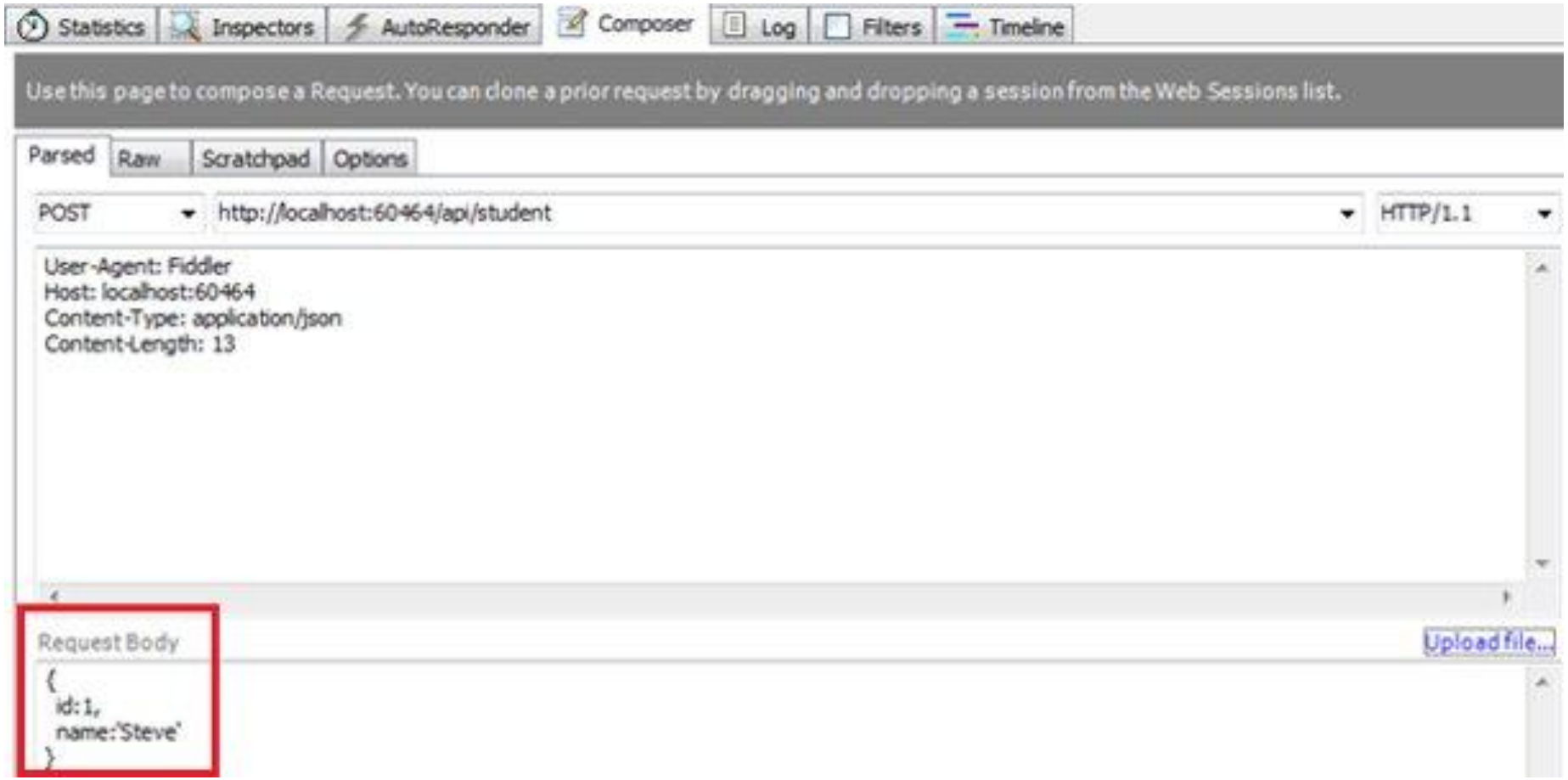
Complex Type Parameter

Example: Post Method with Complex Type Parameter

```
public class Student {
public int Id { get; set; }
public string Name { get; set; }
}
[ApiController]

public class StudentController : ControllerBase
{
public Student Post(Student stud)
{

}
}
```


Statistics
Inspectors
AutoResponder
Composer
Log
Filters
Timeline
Use this page to compose a Request. You can clone a prior request by dragging and dropping a session from the Web Sessions list.
Parsed
Raw
Scratchpad
Options
POST
http://localhost:60464/api/student
HTTP/1.1
User-Agent: Fiddler
Host: localhost:60464
Content-Type: application/json
Content-Length: 13
Request Body
Upload file...
{
id:1,
name:'Steve'
}

[FromUri] and [FromBody]

- You have seen that by default Web API gets the value of a primitive parameter from the query string and complex type parameter from the request body. But, what if we want to change this default behaviour?
- Use [FromUri] attribute to force Web API to get the value of complex type from the query string and [FromBody] attribute to get the value of primitive type from the request body, opposite to the default rules.

To Pass complex object through QueryString

Example: FormUri

```
[ApiController]

public class StudentController : ControllerBase
{
public Student Get([FromUri] Student stud)
{

}
}
```

http://localhost:xxxx/api/student?id=1&name=steve

To Pass simple value through RequestBody

Example: FromBody

```
[ApiController]
public class StudentController : ControllerBase {

public Student Post([FromBody]string name)
{

}
}
```



References

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